

S(i) ranges from -1 to 1

+1 → The point i is well separated.

0 → The point lies on the boundary between clusters.

-1 → The point maybe in wrong cluster.

Agglomerative Clustering

• It's a hierarchical clustering algorithm.

↳ It creates a tree like structure of nested clusters.

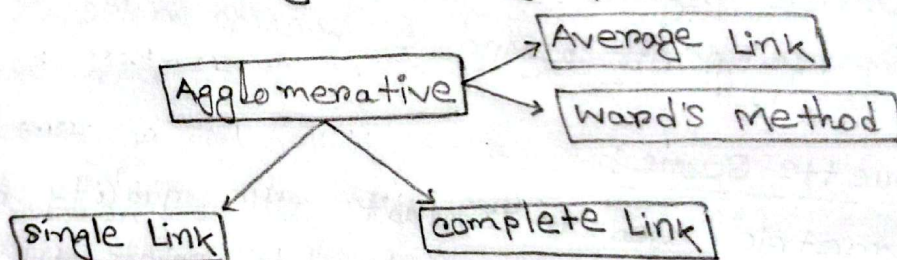
Agglomerative (bottom-up)

start → each point is its own cluster.

step by step merge most similar clusters into a large cluster.

End → all points form a big cluster.

This produces a hierarchy of merges (visualize with Dendrogram)



Single Link (Nearest Neighbor Linkage)

This method defines the distance between 2 clusters as the minimum distance between any point in one cluster and any point in other cluster.

If cluster A has points $\{a_1, a_2 \dots a_n\}$ and cluster B has points $\{b_1, b_2 \dots b_n\}$ then

$$D_{\text{single}}(A, B) = \min_{i, j} d(a_i, b_j)$$

distance between a_i and b_j

Sample

$(0.4, 0.53)$ $(0.22, 0.38)$ $(0.35, 0.32)$ $(0.26, 0.19)$ $(0.08, 0.41)$
 P1 P2 P3 P4 P5

$(0.45, 0.30)$ * all points are a cluster now.
 P6

Step-1 compute distance between every cluster and store known as Proximity matrix

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.14	0			
P4	0.37	0.19	0.13	0		
P5	0.34	0.14	0.28	0.23	0	
P6	0.24	0.24	0.10	0.22	0.39	0

Step-2 merging the two closest members of the two clusters and find the minimum element in the matrix.

0.10 is the minimum so combine P3 and P6

	P1	P2	P3P6	P4	P5
P1	0				
P2	0.23	0			
P3P6	0.22	0.14	0		
P4	0.37	0.19	0.13	0	
P5	0.34	0.14	0.28	0.23	0

$\min((P1, P3), (P1, P6)) \rightarrow P1$
 $\min((P2, P3), (P2, P6)) \rightarrow P2$
 $\min((P4, P3), (P4, P6)) \rightarrow P4$
 $\min((P5, P3), (P5, P6)) \rightarrow P6$

Repeat this process until we got only one cluster.

Agglomerative Hierarchical clustering

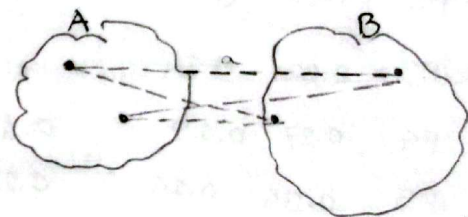
Agglomerative is bottom-up hierarchical clustering

- ① Start: each point = its own cluster.
- ② Steps: iteratively merge clusters based on a chosen linkage criterion.
- ③ End: merge till we got one single large cluster.

① Single Link (Nearest Neighbor)

Distance between 2 clusters = minimum distance between any pair of points (one from each)

$$D_{\text{single}}(A, B) = \min_{i, j} d(a_i, b_j)$$



How it works

At each iteration find 2 clusters which one pair is minimum among all clusters all points.
And merge them into one cluster.
Repeat this until all cluster merge.

If A and B merged then
value of AB = min of all 4 distance.

- Pros**
- . Find non-spherical clusters
 - . Handles irregular shapes
 - . Good when data is naturally connects like a graph.

- cons**
- . chaining effect: may link clusters via thin bridges of points.
 - . Sensitive to outliers (an outlier near another cluster may cause unwanted merging).

② Complete Link (Farthest Neighbors)

Distance between 2 clusters = maximum distance between any pair of point (one from each cluster)

① At start: Each point is a cluster.

② Define distance between A and B cluster

$$D_{\text{complete}}(A, B) = \max_{i, j} d(A_i, B_j)$$

③ At each iteration

- compute all pairwise cluster distances.
- For clusters A, B, C, find values like $D(A, B)$, $D(A, C)$, $D(B, C)$
- Choose the pair with smallest distance.
- merge into a new cluster.

④ Repeat until all clusters merge.

Pros • Produces tight, compact clusters (good for separation)

- Less sensitive to chaining.

cons • sensitive to outliers (cause farthest point dominate)

- Can break large natural clusters into smaller compact one.

When to use

- use when we need well-separated compact clusters.
- common in document clustering.

③ Average Link (unweighted pair group method with arithmetic mean)

Distance between two clusters = avg distance between all pairs of points (one from each cluster)

$$D_{\text{avg}}(A, B) = \frac{1}{|A||B|} \sum_{i \in A} \sum_{j \in B} d(a_i, b_j)$$

How it works

- at each iteration calculate avg distance between all clusters.
- merge 2 with lowest distance.

Pros . Balance between single (too loose) and complete (too strict).

- more robust to outliers.
- Produces more natural groups.

cons . computationally expensive
• not perfect for different densities.

when to use

- avg clustering in between single and complete.
- common in biology (genetic), customer segmentation

④ Ward's Method:

Distance between clusters = increase in total within cluster variance after merging.

- minimizes the sum of squared error (SSE) within cluster.

$$D_{\text{ward}}(A, B) = \Delta \text{SSE}(A \cup B)$$

How it works:

At each iteration

- For every possible merge, calculate how much variance increases.
- Choose the merge with the smallest increase in variance.

Pros . Produces compact spherical clusters like k-means.
• very effective when clusters Gaussian-like.
• Less sensitive to chain and outliers.

Cons . assume that clusters are roughly spherical and similar size.
• Not good for irregular shape.

When to use

- want k-mean type clusters but in hierarchical order.
- very popular in image and market segmentation.